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On the prediction models for brain signal-based emotion recognition

M. Jehosheba Margaret[1], N. M. Masoodhu Banu[2]✉, S. Madhumithaa[2] & Al-Sakib Khan Pathan[3]

This paper analyzes the factors contributing to the performance gap between subject-dependent and subject-independent EEG-based emotion recognition systems. With this key objective, the experiments are conducted in this work using two benchmark datasets; DEAP and SEED across classical machine learning algorithms, KNN, Random Forest, Linear Discriminant Analysis and a deep learning model like CNN and EEGNet. In cross-subject evaluations, CNN achieved the highest accuracy of 84% on the SEED dataset. To interpret these outcomes, Explainable Artificial Intelligence (XAI) techniques using SHAP (SHapley Additive exPlanations) values are employed to identify the most influential EEG features and frequency bands. This analysis not only highlights the key contributors to model decisions but also exposes feature-level inconsistencies that explain the reduced performance in subject-independent scenarios. This diagnostic study provides valuable insights into the limitations of current approaches and highlights the importance of explainability for developing robust EEG-based emotion recognition systems.

Keywords Emotion recognition, Subject independent, BCI, EEG, SHAP

Emotions are the expressions of a human being that are generated based on the perception of external stimuli. Interestingly, the same type of emotion is often different from person to person since it depends on the experience, personality and their emotional intelligence. This includes how well they identify, understand and manage emotions ranging from mild intensity to overwhelming. Other contributing factors that influence variations in emotion are mental health, biological factors, context of the environment, cultural difference and relationships. Due to the complex nature of emotion, there is a difficulty in recognizing the emotions accurately. Emotion Recognition is a promising technique that comes under the broad scope of Affective Computing and Brain Computer Interface (BCI) that helps understand human emotions by interfacing the computer and human brain. There is a high potential to provide improved diagnosis in mental health using this technique.

The acquisition of the signal can be multimodal, i.e., any combination of physiological signals like Electroencephalogram (EEG), Galvanic Skin Response (GSR), Electrocardiogram (ECG), Electromyogram (EMG), Electrooculogram (EOG), Respiration Rate, and Photoplethysmogram and non-physiological signals like speech, text, facial expressions and body gestures. Physiological signals cannot be duplicated as these signals offer higher accuracy and reliability when compared with non-physiological signal. Among them, the most popular method for emotion detection is EEG since it can non-invasively record the electrical activity of the brain. However, there is a serious concern in the emotion recognition system that the EEG data has poor generalization capability across various subjects due to its variability in emotional expression. In such situations, subject independent emotion recognition systems can help recognize emotions despite different individuals, and features across diverse population. Therefore, such systems are preferred for the development of accurate and reliable emotion recognition.

Most subject-dependent models that are trained and tested on the same set of subjects, leads to generalization issues making the model difficult to predict emotions in real time. An analysis of subject independent models is carried out to identify different strategies that could contribute real-time applications. Though there are several articles in the existing literature (some of which will be covered in the subsequent section) that discuss subject independent emotion recognition, there is no clear explanation about the cause of achieving low percentage of classification accuracy in Machine Learning (ML) and Deep Learning (DL) models. The existing models lack interpretability due to the nature of the blackbox model which does not give inner working nature of the model. Hence, in order to understand the underlying issues, Explainable AI (XAI) is used in this article. Thereby, it increases the transparency and credibility of decision-making process of the blackbox model.

[1]Easwari Engineering College, Chennai, India. [2]Vel Tech Rangarajan Dr Sagunthala R and D Institute of Science and Technology, Avadi, Chennai, India. [3]Department of Computer Science and Engineering, United International University, Dhaka 1212, Bangladesh. ✉email: drmasoodhubanu@veltech.edu.in

Image data can be generalized better than EEG dataset due to its inherent characteristics like it is more user-specific and considered a stationary signal[1] whereas EEG signals are more non-stationary and not user-specific. Due to this, there is always a terminology called subject-dependent and independent model in predictions using EEG signal. Applying this concept to EEG-based emotion prediction, most of the researchers found that the model which would be working with inputs of the same group of users would not work with relatively newer user input(s). To better understand the factors behind this behavior, i.e., why models succeed in some cases and fail in others, the proposed work employed decision-support techniques such as explainable AI (XAI) to gain deeper insights into the underlying patterns and model decisions.

The rest of the paper is organized as follows: "Related works" section presents the related works both in subject independent models and the application of XAI in emotion recognition domain. "Overview of explainable AI and motivation" section briefly discusses XAI techniques, "Methodology" section details the methodology followed in the proposed work. "Results and discussions" section presents the results and discussions while "Conclusions" section concludes the work.

Related works

Recent studies emphasize the focus on subject independent models and XAI techniques to enhance the generalizability and interpretability of the model across diverse users. XAI is particularly essential in the healthcare field to support accurate decision-making and to gain the trust and approval of clinicians. This section summarizes the results obtained by various researchers in emotion recognition and other healthcare area.

Subject independent models

Several studies have investigated subject-independent (cross-subject) EEG-based emotion recognition, primarily with the aim of improving generalization across individuals. These works adopt diverse strategies including signal decomposition, feature engineering, architectural innovations, and normalization techniques to address inter-subject variability.

Pandey and Seeja[2] utilized Variational Mode Decomposition (VMD) combined with deep learning to extract frequency-aware representations from EEG signals, demonstrating improved subject-independent emotion recognition on the DEAP dataset. Li et al.[3] conducted an extensive analysis of EEG feature representations under cross-subject settings, highlighting that performance degradation largely stems from inter-subject variability rather than classifier limitations.

Several approaches focus on feature optimization and fusion. Zuo et al.[4] proposed fused entropy-based EEG features, showing that combining multiple entropy measures improves cross-subject robustness. However, the model did not generalize well when tested on an independent dataset, potentially due to demographic and distributional differences, since the SEED dataset is derived from a specific population. Su et al.[5] introduced a genetically optimized projection dictionary pair learning framework, emphasizing optimization-driven feature projection for subject-independent recognition.

Deep learning–based solutions have also been explored. Arjun et al.[6] employed attention-driven neural networks to selectively emphasize discriminative EEG regions, while Ding et al.[7] proposed TSception, a convolutional architecture designed to capture temporal dynamics and spatial asymmetry. Additionally, Fdez et al.[8] addressed inter-subject distribution shifts through stratified normalization, and Ramaswamy and Palaniswamy[9] presented a comparative study using EEG and physiological signals, highlighting the benefits and challenges of multimodal emotion recognition. Since the reviewed studies report accuracies under different classification settings (binary, three-class, and multi-class), Table 1 provides a consolidated summary of their reported performances.

The challenges associated with cross-subject emotion recognition are further exacerbated in depression recognition tasks, where inter-individual variability, heterogeneous symptom progression, and multimodal data acquisition significantly affect model robustness. Comprehensive reviews have highlighted these limitations across diverse populations and sensing modalities. In particular, Yasin et al.[10] presented an extensive survey of machine learning approaches for clinical and non-clinical depression recognition using EEG and audiovisual data. Their analysis emphasized substantial variability across subjects, recording conditions, and relapse stages, and pointed out that most existing studies primarily focus on accuracy-driven evaluation. The review underscores the need for robust generalization strategies and deeper diagnostic analysis, thereby motivating the development of adaptation-oriented and interpretable learning frameworks.

To address cross-subject and cross-domain variability, recent studies have explored domain-adaptation-based deep learning frameworks. Shen et al.[11] have introduced UA-DAAN an uncertainty-aware adversarial adaptation strategy to improve EEG-based depression recognition across subjects, explicitly modelling domain uncertainty. Similarly, Shen et al.[12] WDANet employed Wasserstein distance–based adversarial learning to align feature distributions between source and target domains. While domain adaptation methods improve generalization, they provide limited insight into the neurophysiological mechanisms underlying model decisions. To address this limitation, several studies have incorporated interpretability directly into model architectures using attention mechanisms and factorization-based representations.

The authors in the work[13] have achieved interpretability through self-attention–based factorization, where attention weights explicitly model and quantify functional connectivity patterns, enabling direct analysis of region-to-region interactions without post-hoc explanation methods. The model[14] achieves intrinsic interpretability by incorporating frequency–channel attention factorization, allowing direct inspection of band-specific and channel-wise contributions aligned with neurophysiological knowledge. In both studies, interpretability is embedded in the model architecture through attention-guided factorization, enabling analysis of learned neural representations rather than post-hoc explanation of predictions. To mitigate the impact of inter-

Ref.	Study	Dataset	Classes	Method	Reported Performance
[2]	Pandey & Seeja (2022)	DEAP	Binary	VMD + Deep Learning	61.25%-Arousal 62.50% -Valence
[3]	Li et al. (2018)	DEEP & SEED	3-class	Feature analysis + ML	DEAP-59% SEED-83%
[4]	Zuo et al. (2022)	SEED Own Dataset	3-class	Fused Entropy Features	70.05%
[5]	Su et al. (2023)	SEED MPED & GAMEEMO	3-class	Optimized Dictionary Learning	SEED-69.8%, MPED-24.11 GAMEEMO-64.3%
[6]	Arjun et al. (2022)	DEAP SEED	3-class	Attention-based NN	65.7% -Valence 69.7% -Arousal 76.7 -SEED
[7]	Ding et al. (2023)	SEED MANHOB-HCI	3-class	TSception(CNN)	61.5%-arousal 59.14% -valence 60.61%-arousal 61.27% -valence
[8]	Fdez et al. (2021)	SEED	3-class	NN + Stratified Normalization	79.6%
[9]	Ramaswamy & Palaniswamy (2022)	DEAP / Physiological	Binary / Multi	Comparative ML/DL	55% to 60%
[15]	Valderrama (2026)	SEED, SEED-FRA, SEED-GER,	3 class	Demography aware attention mechanism	80.2%, 80.5%, 88.8%
[16]	Valderrama (2025)	SEED, SEED-IV, SEED-V	3 class	Attention Network	79.3% 60% 60.7%

Table 1. Summary of EEG based emotion recognition studies.

subject variability, several studies have incorporated interpretable architectural constraints, such as attention mechanisms and factorization frameworks, to explicitly model shared neural representations across subjects.

While attention mechanisms have been widely adopted in diverse EEG applications to enhance feature discrimination and model interpretability, their role in addressing subject variability in EEG-based emotion recognition has only recently gained attention. In this direction, Valderrama et al.[15] examined the impact of demographic factors on subject-independent EEG emotion recognition by fusing EEG features with demographic attributes through an attention-based framework, demonstrating improved cross-subject generalization on SEED-derived datasets. Similarly, Valderrama and Sheoran[16] employed spatial and temporal attention layers to identify EEG channels that consistently contribute to emotion recognition across subjects, revealing that a reduced subset of channels dominates subject-independent performance.

XAI in EEG based emotion studies

Some of the researchers, through XAI, suggest that it is possible to improve the generalization of EEG-based emotion recognition models by understanding the features that contribute to predictions across different subjects. This understanding helps to develop more robust models, which can adapt to new subjects or environments. They used saliency, and guided back propagation method as XAI and applied the technique exclusively to SEED dataset. An interesting work carried out in investigated how different frequency bands contribute to emotion recognition; however, the explanations are limited. They performed the experiment on Dreamer dataset. Alharbi used multimodal emotion recognition dataset constructed under virtual reality environment. The experimentation was carried out by taking 44 features for both machine learning and deep learning models. The authors used summary plot of SHAP to visualize the most important and contributing feature for making decisions. Fiorini et al. used integrated gradient method of XAI to interpret the prediction of the model. The EEG data were clustered into time and channel groups, and analyzed using linear mixed-effects models for different architectures. Among the various architectures evaluated, the transformer model demonstrated a distinct pattern in interpretability analysis using the Integrated Gradients method and achieved the highest accuracy overall. Mouazen et al. used SHAP to understand the decision making process of the model on DEAP dataset. The summary plot showed the distribution of features for each frequency band. Among them, Gamma band has shown relatively higher density in feature distribution.

Though there are different deep learning techniques to address the subject independent emotion recognition on DEAP and SEED dataset, their overall performance remains inconsistent (often significantly lower on DEAP). This suggests underlying challenges in generalizing across subjects, which is due to complex individual-specific neural patterns and dataset-specific characteristics. Although some recent studies have incorporated XAI techniques to interpret model decisions, the insights provided have often been limited or lacked sufficient experimental validation. In order to address this gap, the proposed study employs SHAP a robust XAI method, as a diagnostic tool to systematically analyze and explain the behavior of subject-independent emotion recognition models, particularly on the DEAP dataset. By uncovering the contribution of features and the variation in learned patterns across individuals and datasets, this diagnostic analysis brings to light the underlying factors responsible for the observed performance differences, while emphasizing the role of explainability in EEG-based emotion recognition.

Motivated by these explainability-driven studies and with the objective of providing a diagnostic analysis of EEG-based emotion recognition models under subject independent and dependent evaluation settings, through

a systematic comparative study using SHAP, enabling deeper insight into model behavior and inter-subject variability, this paper adopts XAI techniques and the key contributions are listed below:

- A comparison is carried out among three different ML frameworks: KNN (K-Nearest Neighbors), LDA (Linear Discriminant Analysis), and RF (Random Forest). Further, it has been extended to DL framework with multiple layers of CNN.
- While subject-dependent models are known to exhibit higher accuracy due to personalized patterns in the data, subject independent models present greater challenges due to inter-subject variability. To address this gap, we analyze the performance disparity between these scenarios.
- Furthermore, to ensure the interpretability of the models, Explainable AI techniques, are employed, specifically SHAP (Shapley Additive Explanations), to uncover feature contributions and provide insights into model decision-making. SHAP values are leveraged to identify key EEG features influencing predictions in both subject-dependent and subject independent contexts, revealing critical patterns that impact model performance.

Overview of explainable AI and motivation

XAI is used to validate the obtained outcome of the subject independent evaluation results by providing a better transparency and understanding of the model's decision process. It helps analyze the strength and weakness of the model and thus aids in improving the model. XAI consists of different set of methods and techniques that help human interpret and understand the Artificial Intelligence (AI) models easily. It provides transparency by informing how the inputs are processed to produce the outputs. It brings out the influence of features in forming the output and model behavior. It also bridges the gap between the complex AI systems and human understanding in various fields. Different techniques are used for analyzing the models of various applications with model-agnostic approaches like SHAP, LIME (Local Interpretable Model-agnostic Explanations), and so on. There are many XAI packages available. Table 2 shows some of the open sources that are available and their explanations in terms of visualization.

The above mentioned tools are used in various medical applications like cancer detection[22], Parkinson's disease detection[23], respiratory disease detection[24], heart disease prediction[25], kidney disease[26], Alzheimer's disease diagnosis[27], Stroke prediction[28], etc. XAI is used for various signal processing applications like speech[29], and radar[30]. It is also used in EEG signal analysis for sleep apnea detection[31], seizure detection[32,33], Event related potential[34], and emotion recognition[17,18]. The authors of[35] provide a comprehensive analysis of various XAI works in various domains by systematically analyzing the literature, highlighting that, there are only limited works available with EEG as a base. By integrating LIME, the authors of[36] identified the contributing features for motor imagery application; however, they did it for each instance as the XAI model is local not global. The study by the authors of[37] with XAI investigates the feature changes due to data poison attack like label changing.

Although these previous works explored and investigated the application of XAI in EEG based application, they lacked the global explanation for each emotional class. These earlier results motivate further research to widen the applications for improving the decision-making systems. This work takes a significant step forward towards developing a BCI based emotion recognition system able to be interpreted by XAI. This work not only identifies the local individual features for each instance but also identifies the influence of global features contributing to each class. Furthermore, a better understanding of the relationships between inputs and outputs provided by the XAI explanations can enable production of more effective EEG acquisition devices or algorithms. Therefore, this paper addresses a XAI (SHAP) framework for getting insights on the influence of EEG features in determining the outcome and aiding in the interpretation of neurological conditions in "Overview of explainable AI and motivation" section.

Methodology

The recognition of emotion requires few steps, so that the EEG signal from the brain can be processed in the machine. The block diagram in Fig. 1 depicts the work flow of emotion recognition and the validation of subject independent system. The EEG signals are collected from the brain and then processed to extract information using frequency domain features. The information is processed in the ML or DL framework for the recognition of

Explainer	Dimension	Visualization	Interpretation
SHAP	1-D and 2-D	SHAP Summary Plot	It gives global view of feature importance and their effects on predictions. It gives the important features as SHAP values.
		SHAP Force Plot	It helps visualize the contribution of each feature for single prediction.
		SHAP Dependence Plot	It shows the effect of single feature in interaction with other features across the dataset.
		SHAP Waterfall Plot	It provides a detailed view of each feature contributing to the final prediction for a single instance.
	1-D 2-D	SHAP Beeswarm Plot	It shows clear view when dealing with multiple features and instances.
		SHAP Image Plot	It helpsvisualize SHAP values based on pixels that contributed most to the prediction in the image data
LIME	1-D	Bar Plot	It shows the importance of each feature for a specific instance.
		Summary Plot	It displays the feature values and their contributions to the prediction
	2-D	Super pixel Highlighting	It shows the superpixels that influence the most for the prediction.
		Heatmaps	It helps visualize the region of importance in the image. It provides correlation between different features.

Table 2. Explainers and their interpretation.

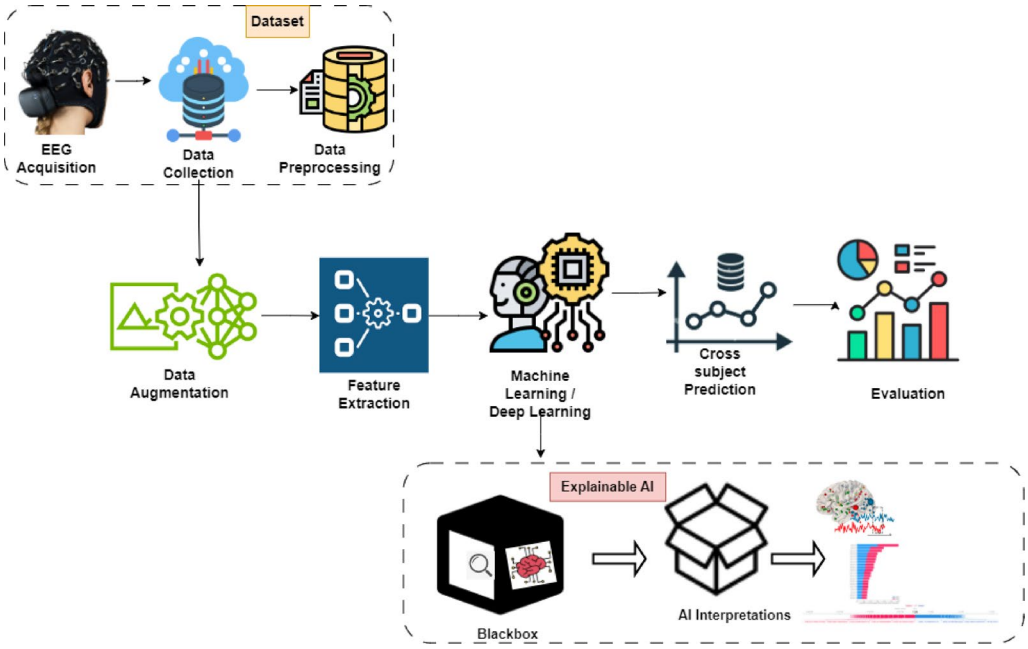


Fig. 1. Block diagram of subject independent EEG based emotion recognition using XAI.

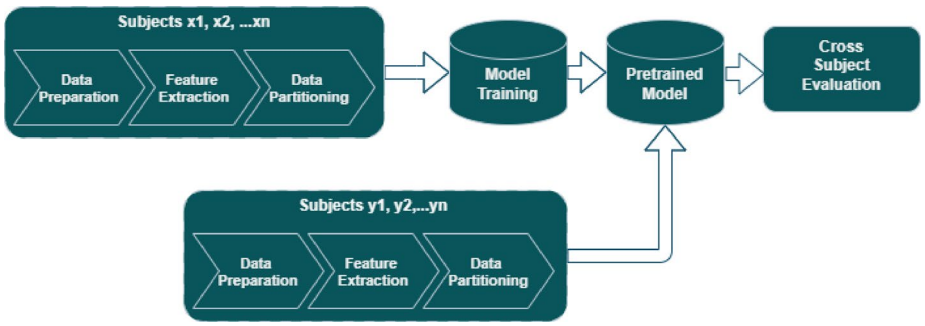


Fig. 2. Subject independent evaluation for emotion recognition.

emotions. After this, the models are cross validated using subject independent system in the dataset. Further, the features contributing to the classification of emotion is highlighted using explainable AI namely SHAP methods.

Subject independent evaluation

Subject independent evaluation is a testing process where the subjects that were not part of the training set are evaluated on the trained model. This testing helps assess how well the model generalizes to the new subjects since emotions tend to vary from person to person. Subject independent evaluation is crucial to ensure robustness and reliability of the developed model for real-time applications. The process of subject independent evaluation is described using the block diagram shown in Fig. 2.

Dataset description

DEAP dataset, which is one of the largest datasets for Emotion Recognition was considered to evaluate the performance of the models for subject independent system. The other dataset used in this experimentation was SEED (SJTU Emotion EEG Dataset). It was collected and was made available for emotion-based research by Shanghai Jiao Tong University. The details of DEAP dataset and SEED dataset are given in Table 3. The structure of data and label for DEAP and SEED datasets are presented in Table 4.

In the preprocessing stage, each of the datasets was downsampled and filtered to remove artifacts present in it. In SEED dataset, artifacts were removed by band-pass filter in the range of 0–70 Hz.This in turn reduces the noise in the signal, reduces the high dimensionality and improves Signal-to-Noise Ratio (SNR).

Data augmentation

The EEG signal in time domain can be represented by the Eq. (1), i.e., multidimensional time series:

Dataset	DEAP	SEED
Physiological experiment		
Number of participants	32	15
Number of videos	40	15
Rating Values	1–9 (continuous scale)	3
Rating Parameter	Valence, Arousal, Dominance, Liking	Positive, Negative and Neutral
Number of channels	32(EEG), 04(EOG), 04(EMG), 04(GSR), Plethysmography, Body temperature and respiration	62
Recording Frequency	512 Hz	1000 Hz
Downsampled frequency	128 Hz	200 Hz
Bandpass filter	4–45 Hz	0–75 Hz

Table 3. Experimental details of SEED dataset.

Name	Structure	Contents
DEAP dataset		
Data	40 × 40 × 8064 (128 Hz x 63 s)	(Videos/Trials) x No. of Channels x data
Labels	40 × 4	(Videos/Trials)x (Valence, Arousal, Dominance, Liking)
SEED dataset		
Data	3 × 15 × 62 × 4800 (200 Hz x 240 s)	Experiments x (Videos/Trials) x No. of Channels x data
Labels	3 × 15 × 3	Experiments x (Videos/Trials) x (Positive, Neutral, Negative)

Table 4. Structure of Data for DEAP and SEED.

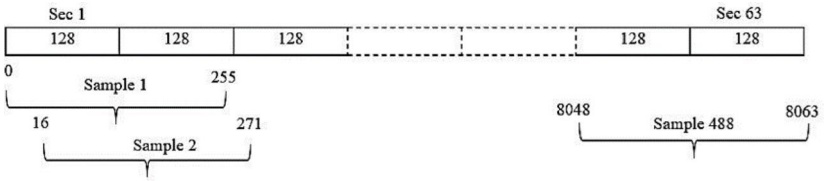


Fig. 3. Data augmentation of DEAP dataset.

$$X_i\,(n) = [x_1, x_2, \ldots\ldots, x_T] \in\ R^{N_X T} \tag{1}$$

where N is the number of channels and T is the number of time steps and $i$is the subject number. The original data without augmentation has $N = 14\ and\ T = 63{*}128$. The data needs to be segmented, as real-time emotion prediction is not possible with such a long-time sequence and these are referred to data chunks. In addition, the DL models need huge amount of data. Hence, in order to increase the number of samples (data chunks) for testing and training, data augmentation is carried out in DEAP dataset. The recorded video of $63\ sec$ is augmented into 488 data chunks with the window size of $2\ sec$ and overlap of $0.125\ sec$.The augmentation of single data sample of 8064 time step is shown in Fig. 3.

In SEED dataset, data was augmented similarly using windowing technique with overlap. Under each experiment, the user emotions were recorded for $15\ trials$. However, the length of the recording is not the same for all trials. Hence, the augmentation result for each subject was different, but can be generalized using the formula (2).

$$Augmented\ Data = \frac{(Length\ of\ the\ trial * Sample\ rate) - Segment\ Length}{(1 - overlap\ percentage) * Segment\ Length} \tag{2}$$

In this case, the total sample size is 36,800, with a window size of 400. For each experiment, 15 trials were conducted. When segmenting the data into non-overlapping chunks, a total of 4,140 data samples/chunks were generated. When using a 50% overlap (i.e., a step size of 200), the number of data chunks increases to 8,280.and are given by Eq. (3) as,

$$DataChunks\,(X_{cni}) = \{X_{11,}, X_{12,}\ \ldots\ldots\ldots\ldots X_{1n,}\} \in\ R^{N_X T} \tag{3}$$

where T, here is 256/400, i.e., 2 sec time window and N is the same 14/64 channels for DEAP and SEED datasets respectively. X_{cni} are of different data chunks derived from the Eq. 1 after augmentation, where cn represents

chunk number from each subject i. $X_{xy,}$, where the suffix x represents the subject number and the suffix y represents time samples within it.

Feature extraction

The raw data chunks are in the dimension 14x256, where 14 is the number of channels and 256 is $2\ sec$ time window of data. In order to reduce the dimensionality, feature based input is considered for analysis. Hence, Fast Fourier Transform (FFT) was applied over the EEG signals to convert them into frequency domain.

$$X_K(f) = \sum_0^{N-1} X_c\, e^{j2\pi \frac{kn}{N}} \tag{4}$$

$$P(f) = |X(f)|^2 \tag{5}$$

where X_c is given by Eq. 3 and $X_K(f)$ is the Fourier coefficient which is in the same dimension of X_c obtained from Eq. 4.

The next step is to get the Power Spectral Density (PSD) features, which were extracted from the five EEG rhythms namely *theta*, *alpha*, *sigma*, *beta*, and *gamma* which would lie between 4 and 30 Hz. In order to optimize the data, 14 channels were selected based on Emotive Epocsystem[38]. The resulting data is $19{,}520\ (augmented\ data\ per\ subject)\ x\ 70\ (14\ channels * 5 bands)$ after applying the formula from the Eq. 5. A detailed explanation of the preparation of data for DEAP dataset could be found in[39]. For SEED dataset, the final data obtained after augmentation and feature extraction without overlap is in the size of $8280\ rows\ x\ 310\ columns\ (62\ channels * 5\ power\ signal\ band)$.

Model architecture

Both ML and DL models classified the extracted features. The ML models that were used for classification were Random Forest, K-Nearest Neighbor (KNN) and Linear Discriminant Analysis (LDA). The ML models that gave good validation prediction in the previous experimental work[40] were selected. Convolutional Neural Network (CNN) was used as the DL model and the layer specification of CNN are shown in Table 5. The features that were extracted by using FFT were converted into single dimension. The activation function used in this model is ELU since it performs well for EEG based emotion recognition[41].

After the classification of emotions based on the number of classes, the results are validated using other performance metric like F1-score, Accuracy, Recall and Precision. The model's performance is predicted using confusion matrix that gives the summary of the predictions made by the model in comparison with the original label of the data.

EEGNet In addition to the above discussed feature-based pipeline, EEGNet—a compact convolutional neural network designed for EEG signal classification—was also tested for fair comparison. Unlike the machine learning classifiers and the 1D CNN that rely on handcrafted and augmented features, EEGNet directly learns spatial–temporal representations drawn from raw EEG signals without explicit feature engineering.

Results and discussions

This section focuses on XAI techniques to enhance the interpretability and transparency of emotion recognition models. It examines the performance of the developed models, highlighting their accuracy. Additionally, it delves into the insights provided by XAI approaches, such as feature importance, decision making, and classification patterns, offering a deeper understanding of the model's behavior. By combining quantitative results with qualitative explanations, this analysis bridges the gap between model performance and interpretability, ensuring the models is both effective and trustworthy for practical applications. Experiment was conducted to compare subject-dependent and subject independent models across 2, 4, and 16 classes. The accuracy of these models was evaluated over ML and DL models. The results are presented in Table 6 for DEAP and in Table 7 for the SEED.

The raw EEG data of each subject were segmented by overlapping of windows as described in Sect. 4.2; each data chunk served as an independent instance. The data chunk and the original data had the same label instance. In this way, the signals in each data chunk were used as instances to train the model. The model trained with window size of 2 s, with the sliding window length of 1.75 s, gave a classification accuracy of 98.7%. This shows that the longer the sliding window's duration, the greater the amount of historical data in the

Layer	Filter/ No. of Neurons	Kernel	Activation	Max Pooling	Dropout	Batch Normalization
Parameters of CNN model						
Conv1D	128	3	Yes	2	–	Yes
Conv1D	128	3	Yes	2	–	Yes
Conv1D	64	3	Yes	2	–	No
Flatten	–	–	–	–	–	–
Dense	64	–	Yes	–	0.2	–
Dense	32	–	Yes	–	0.2	–
Dense	16	–	Yes	–	0.2	–
Dense	4	–	Softmax	–	–	–

Table 5. Parameters of CNN model.

Model	Class	Validation accuracy (1–10 sub)	Cross validation-1 (21-25sub)	Cross validation – 2 (11-15sub)
Random Forest	2	94.68	54.04	50.07
	4	92.16	25.34	26.14
	16	91.61	9.174	8.785
KNN($N=7$)	2	94.50	51.26	50.42
	4	90.66	25.31	31.95
	16	88.13	9.58	11.90
LDA	2	69.88	51.91	54.70
	4	41.64	21.32	20.43
	16	25.19	05.03	03.14
CNN	2	93.68	57	57
	4	90.05	24.6	26.2
	16	84.84	13	14
EEGNet	4	66.17	47.7	45.1

Table 6. Accuracy of subject-dependent and independent models for DEAP dataset.

Model	Class	Training accuracy	Subject independent accuracy in %
RF	3	100	75
KNN	3	83	45.1
LDA	3	61	58
CNN	3	92	84
EEGNet	3	63.9	33.01

Table 7. Accuracy of subject dependent and independent models for SEED dataset.

window, which contributes to better classification accuracy. Under identical experimental settings, except for the absence of explicit feature extraction, EEGNet failed to achieve comparable performance on both the SEED and DEAP datasets. This behavior can be attributed to its architectural emphasis on spatial filtering rather than temporal feature learning. A statistical study also has been conducted to evaluate the superiority of the CNN model with EEGNet and to support further evaluation. The CNN model demonstrates consistent performance improvements over baseline methods; however, the difference did not reach statistical significance ($p=0.125$), likely due to the limited sample size. The large effect size (Cliff's $\delta=1.0$) and the narrow confidence intervals indicate stable and practically meaningful performance across subjects.

However, with the same, approach, when the model was used for subject independent prediction as described in Sect. 3, the classification accuracy decreased. Also, it can be observed from the Table 6 that DEAP dataset shows suboptimal performance in subject independent model for any of the ML and DL models whereas SEED dataset gave an improved accuracy of 84% with CNN. This peculiar behavior led to this preliminary study that, SHAP, an explainable AI tool was used to diagnose the reason behind the model behavior. Since SEED showed good performance on subject independent setting with CNN, it was additionally tested for cross dataset condition, i.e., DEAP dataset was tested on SEED trained CNN model. However, the issue here was the number of channels. Hence, the SEED data also cut down to 14 channels similar to DEAP in accordance with Emotive headset[38]specific channel locations and their equivalency is given in the below Table 8.

As a part of cross dataset validation, the SEED dataset trained model was evaluated on DEAP dataset. Hence, the SEED data was aligned to DEAP dataset in training for compatibility. Specifically, the channel was reduced to 14 channels as described above and the emotions are cut short to two removing neutral in order to maintain the classification consistency. The prediction accuracy observed was 88.2ensuring no ambiguity with the participants when number of emotion classes are less.

SHAP has methods for both local and global interpretability. Global explanation approaches provide an explanation that describes the all-inclusive logic of the entire model for any input instance, i.e., it returns a generalized explanation for the decisions that are valid for the whole set of the input X (19520 data samples for DEAP and 8280 data samples for SEED). Alternatively, the local explanation approaches explain the behavior of a model for a specific instance, i.e., it unmasks the reasons for the classification only for a specific instance within the set X. SHAP works on the principle SHAP values given by:

$$\varnothing_j(f) = \sum_{S \subseteq \mathrm{N} \setminus \{j\}}^{n} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} [f(S \cup \{j\}) - f(S)] \tag{6}$$

Model sensitivity and ablation analysis

To systematically evaluate the contribution of individual components to the overall model performance, an ablation study was conducted. To ensure fair and meaningful comparisons, the ablation study was performed only

14 Channel locations	DEAP Dataset	SEED Dataset
AF3	1	4
F7	2	5
F3	3	6
FC5	4	8
T7	6	12
P7	11	14
O1	13	16
O2	17	22
P8	19	24
T8	20	32
FC6	21	42
F4	25	50
F8	29	59
AF4	31	61

Table 8. Channel correspondence between DEAP and SEED Datasets.

Layer number	Model 1	Model 2	Proposed model
CNN layer 1	64	128	128
CNN Layer 2	64	64	128
CNN Layer 3	32	32	64
Accuracy in %	81.33	80.78	83.21

Table 9. Accuracy Variation with number of filters in CNN Architecture or SEED Dataset.

Model	Dropout value	Accuracy in %
Model 1	0.4	78.82
Model 2	0.1	81.53
Proposed model	0.2	83.21

Table 10. Accuracy variation with dropout ratio in CNN architecture for SEED dataset.

Model	Step size in samples	Accuracy	Window length in secs	Accuracy	Number of channels	Accuracy
Model 1	32	82	1	66.6	32	92
Proposed model	16	91.2	2	91.2	14	91.2
	16	92.1	3	92.1	10	81.5

Table 11. Accuracy variation with step size and window length for DEAP dataset.

on parameters that are common to both datasets, namely the hyperparameters. Channel and feature ablations were not included because channel configurations differ substantially across datasets (SEED vs. DEAP) and feature representation differs across the tested model architectures (CNN vs. EEGNet), making such ablations not directly comparable. Hence, the key hyperparameters were selectively varied and the resulting changes in classification performance were analysed to assess the effectiveness of each component. Specifically, the ablation experiments focused on the number of convolutional filters and the dropout ratio. These parameters directly influence the model's representational capacity and regularization behavior and is shown in the Tables 9 and 10.

The hyperparameters of the proposed architecture were optimized first under a fixed signal configuration (2s window) and 14 channel on SEED dataset. To maintain experimental feasibility, the ablation study on signal parameters were tried on DEAP dataset The alternative window durations, step size and number of channels while keeping architecture fixed were studied to ensure controlled comparison and are tabulated in Table 11. On keeping the window duration as 2 s, a step size of 16 and 32 was evaluated. It was observed that, the accuracy decreases as the step size increases, which can be attributed to a fewer training samples and minimal temporal connectivity due to reduced overlap. Next with the step size fixed at 16, different window duration was evaluated. It was found that higher window duration up to 4 s gave higher accuracy with slight overfitting. Hence, 2 s window duration was fixed as a final model parameter and the final model settings are tabulated in Table 12.

CNN layer-1	CNN layer-2	CNN layer 3	Dropout ratio	Step size	Window duration	Bands
64	64	32	0.2	16	2 s	α, β, θ § and γ.

Table 12. The final model parameters.

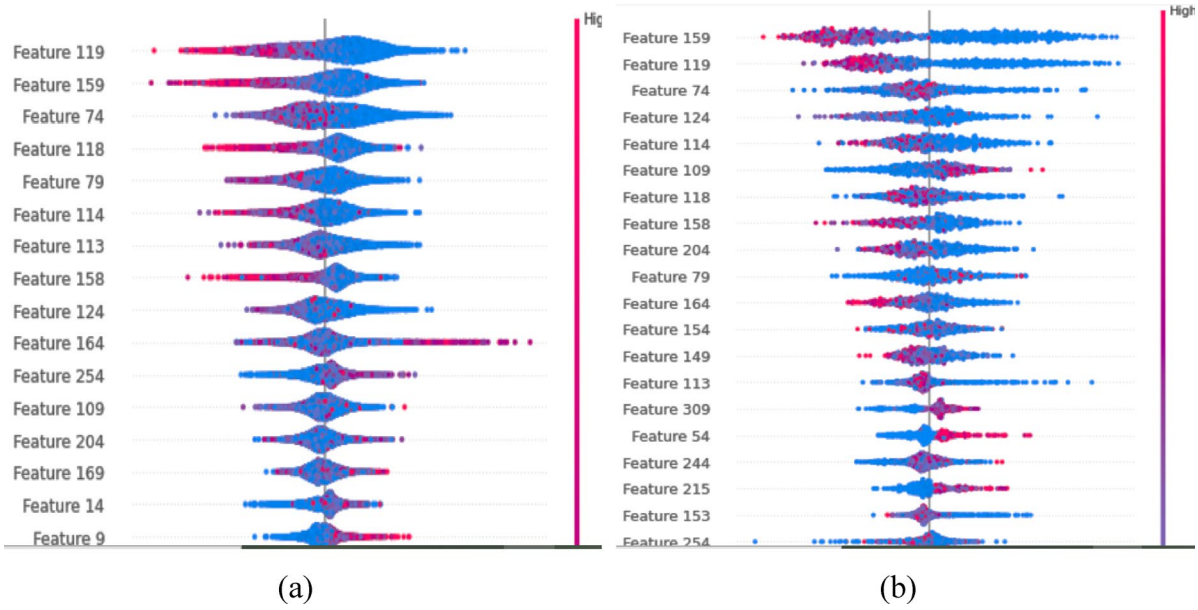


Fig. 4. SHAP value impact on model output: (**a**) subject-dependent, (**b**) subject independent.

Once the hyperparameters and segmentation parameters were fixed, the ablation study was conducted on frequency bands by evaluating the different subsets of EEG bands. Out of five bands relevant to EEG three bands contributes more strongly to emotion and hence the ablation study was done with all the bands and the three bands which are α,β and γ. Compared to all the bands, the three bands configuration gave an accuracy of 82.5 which is less than all bands configuration. It is because θ band also contributes to emotion in some cases.

Subject independent emotion classification

There are multiple visualization plots available with SHAP, which are good on interpretation in their own way. For example, SHAP decision plots show how complex models arrive at their predictions for an instance whereas the bee swarm identifies the most important features in the model in their order of importance.

The plots shown in Fig. 4a and b show the bee swarm plots for the SEED dataset with different accuracies for subject-dependent and independent scenarios trained with Random Forest ML model with the classification accuracy of 0.75 and 0.572 respectively. The reduction in accuracy with subject-dependent model can be explained with SHAP Bee Swarm plots as it represents global explanation. Each dot in the above Fig. 4 represents the SHAP value of a particular feature for one data point. The position of the dot along the x-axis indicates the magnitude and direction of the feature's impact on the prediction for that instance. The number of dots (data points) for each feature shows how many instances that feature affected. A larger number of dots indicate the feature's impact was observed in more instances. Hence, it can be concluded that; Fig. 4a features have a dense feature spread than the Fig. 4b features which are scattered, which is the reason for its high accuracy. The feature values of the model for any classification are not unique or not at least a group of values, instead the spread is too vast that at some instant, it fails to predict correctly and hence, exhibits low accuracy.

Understanding discriminative features for different emotions

The same Bee Swarm plot for different classes can be used to find the important and discriminative features that are used to predict the classes correctly. Figure 5 shows the three bee swarm plots of different classes like positive emotion, negative emotion and neutral class. Here, the red arrows represent feature effects, i.e., the SHAP values that drive the prediction value higher while blue arrows are those effects that drive the prediction value lower. Each arrow's size represents the magnitude of the corresponding feature's effect. If the top 3 to 4 common features for all the three classes are considered, it can be listed as 159,119,118, and 114.

It can be seen from the Fig. 5 that the SHAP values in the negative range are positively influencing to predict the positive emotion, i.e., the red points are on the left side and the blue points are on the right side for the features 159 and 114. More are less similar in case of neutral emotion also; however, the red and blue dot SHAP values are completely opposite for the negative emotion. This explanation seems to be correct as the feature cannot be in the same magnitude and direction for two different opposite emotions. Sometimes there may be

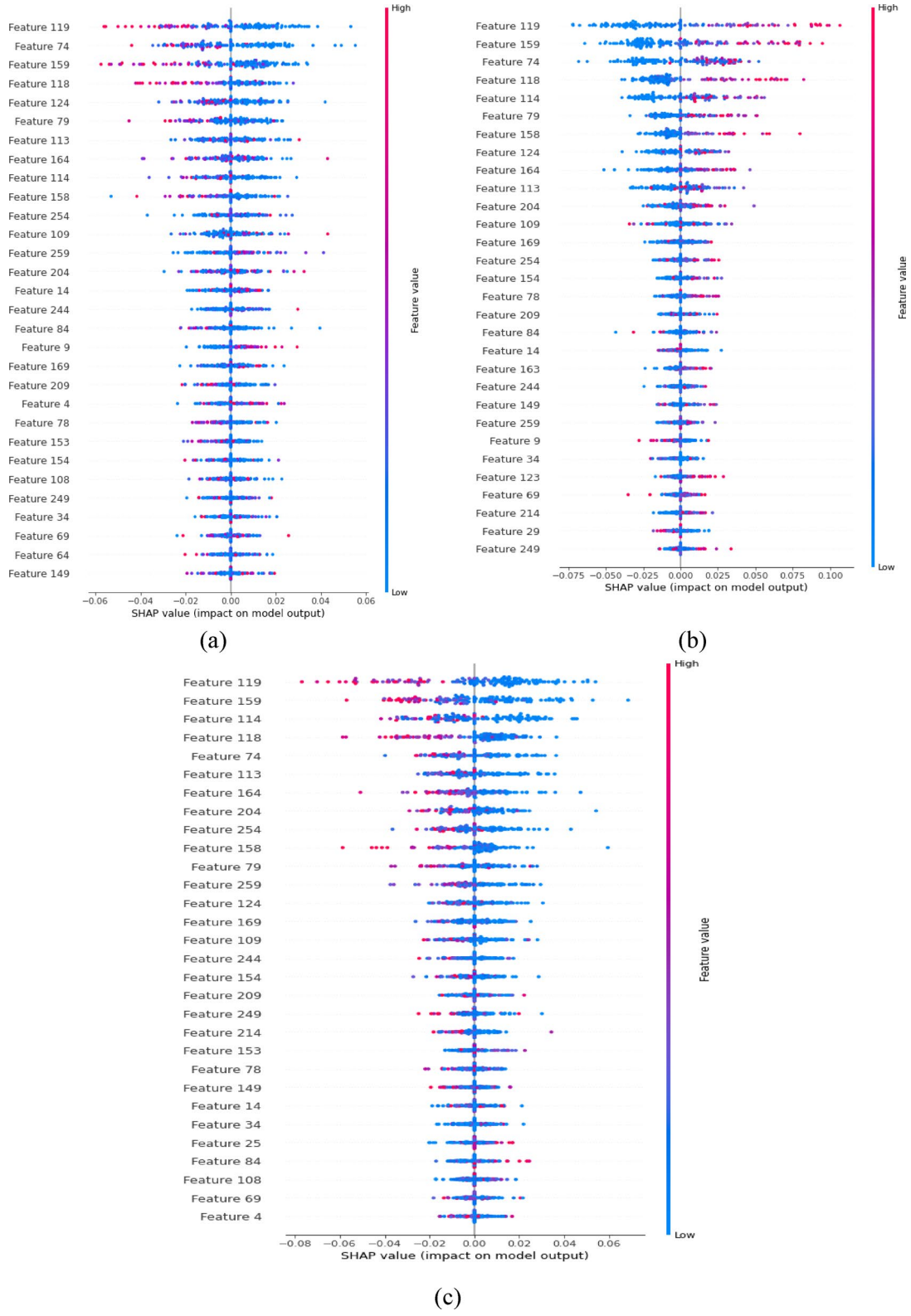


Fig. 5. Bee Swarm plots of different classes for case 1- classification accuracy 0.572: (**a**) positive emotion, (**b**) negative emotion and (**c**) neutral class.

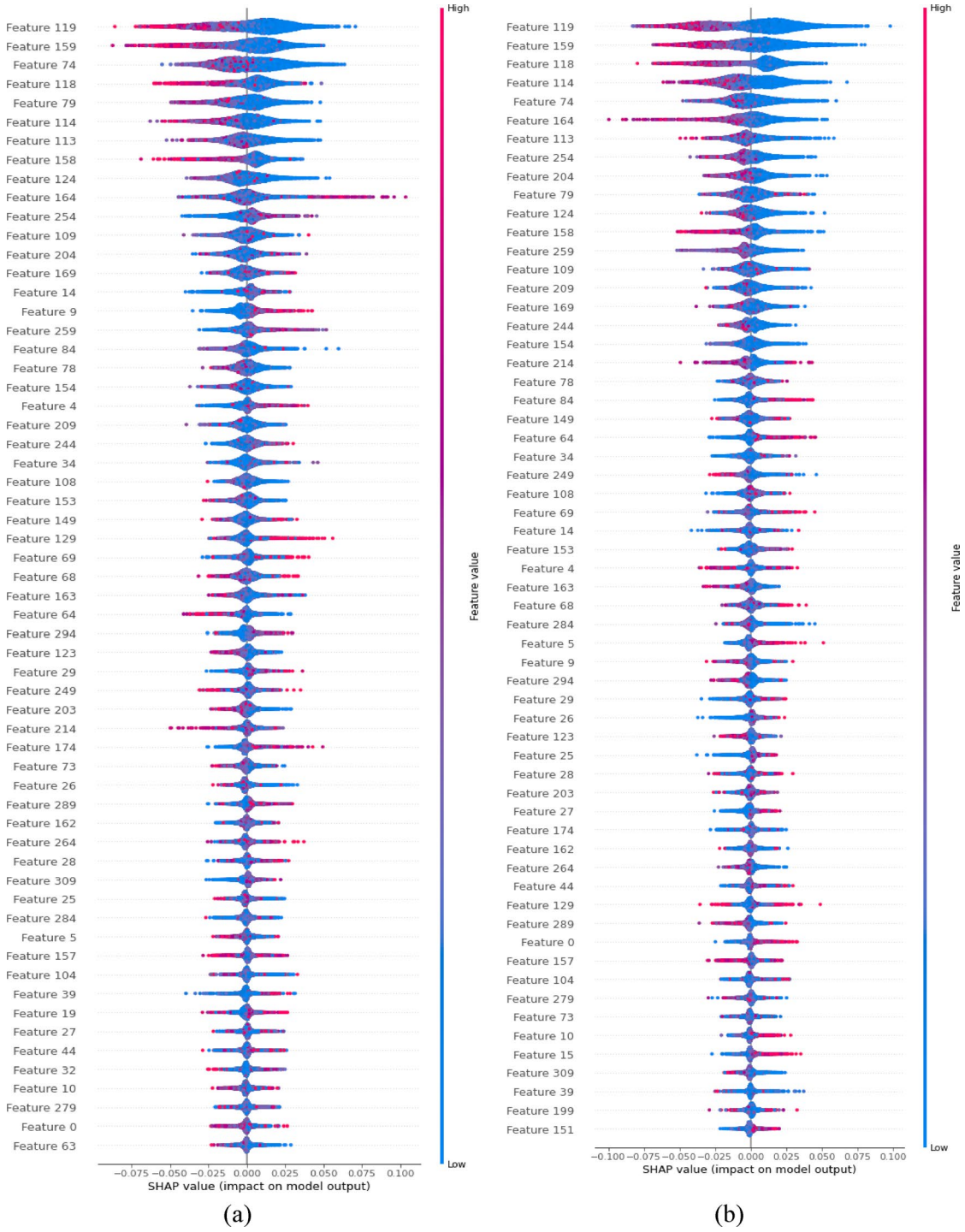
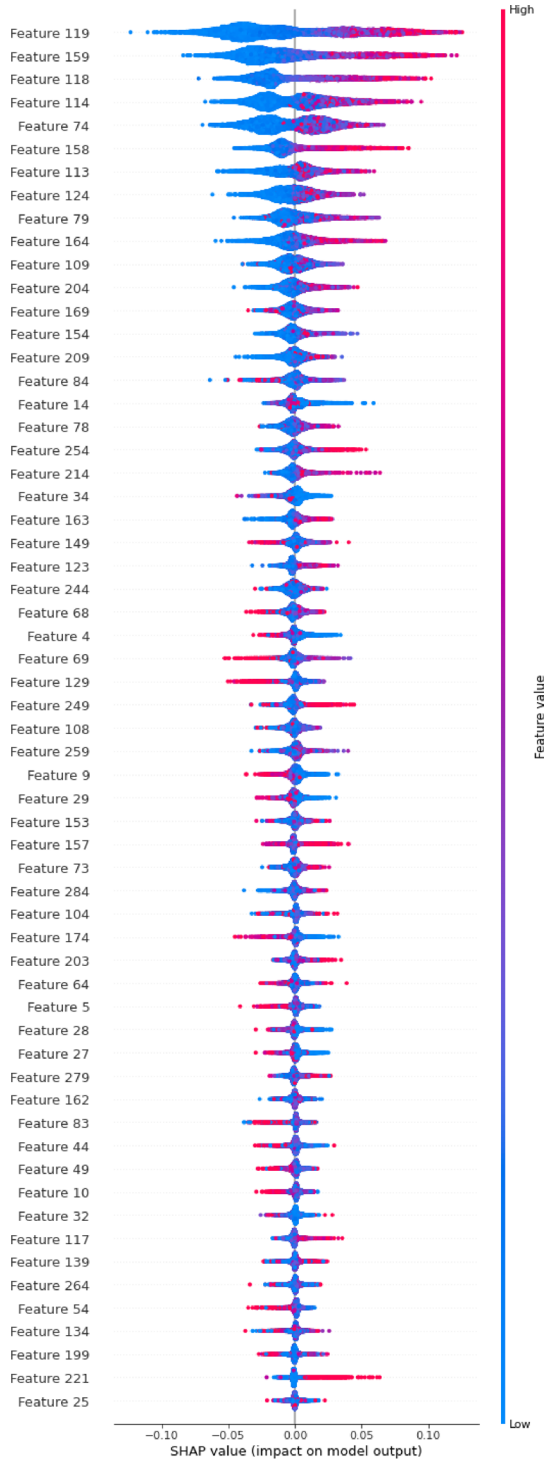


Fig. 6. Bee Swarm plots of different classes for case 1- classification accuracy 0.75 (**a**) positive emotion, (**b**) neutral emotion and (**c**) negative class.

outlier points also, i.e., in the feature 119, some blue points are lying on left side, where major points are red. Most of the features show this characteristic between positive and negative emotions; however, some features like 244 are completely blue in color for both the cases. Between positive and neutral class, there is no information discriminative features. Feature characteristics could be drawn as feature direction for red and blue points, which are the same as well (of the magnitude), not clearly differentiable between these two classes.

On comparing Fig. 5 with Fig. 6, which corresponds to classification accuracy 0.75, it can be deduced that, for most of the features irrespective of their SHAP magnitudes and direction, there is a definite positive and negative component except for a very few features, whereas in Fig. 5, most of the features are not discriminative. For example, the features 74, 78, 124 and 169 are almost the same for both positive and negative emotions in Fig. 5



(c)

Fig. 6. (continued)

whereas some differences can be seen with Fig. 6.This performance degradation on DEAP serves as a diagnostic indicator of the model's inability to handle emotional ambiguity and uncertainty inherent in continuous valence–arousal labels (DEAP dataset).

Understanding the single instance prediction with decision plot

Case-1: right prediction

Single instance or local interpretation can be done either by force plot or decision plot. The decision plot is of two types: single-output and multioutput, where the multioutput plot can be used for multiple class classification. For

right prediction, SEED dataset experiment is considered as its classification accuracy is 0.75 and hence, most of its predictions would be correct. The total number of classes for classification here is three and they are positive, neutral and negative classes.

Figure 7 shows the multioutput decision plot for the instances: 14,789 and 250. Decision plot shows the predicted output in red color for two instances in Fig. 7a, b and c and the class predicted is class 0, class 1 and class 2, respectively. Also, it shows how the individual features are added up to decide the final predicted output. It was confirmed that, the particular instance predicted through SHAP was class 1 by manual checking, i.e., by identifying the ground truth as shown in Table 13.

Also, it can be seen that though the global features list from bee swarm plot is almost the same as each instance decision plot, the order of importance with respect to SHAP magnitude is different. This is also one of the reasons for subject independent accuracy reduction. It means that the 5 band features from the 62 channels are different for different individuals. If it is unique, among subjects, the global importance and local (instance) features will be same and hence, subject independent accuracy can be increased.

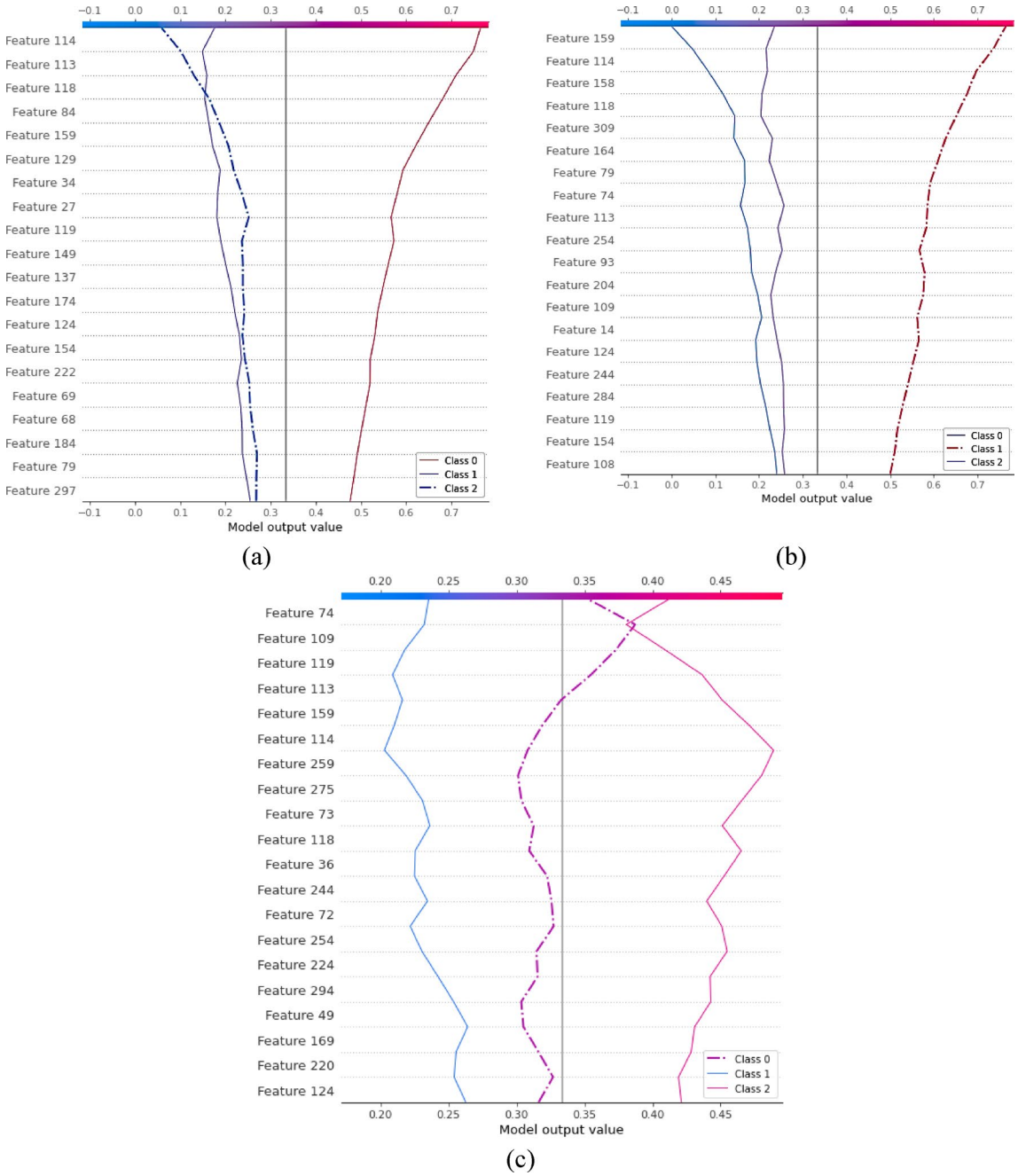


Fig. 7. Multioutput decision Plot of SEED: (**a**) for Class 0 - Instance 14, (**b**) for Class 1- Instance 789, (**c**) for Class 2 -Instance 250.

Instance number	Ground truth label	Predicted label from RF algorithm	Predicted label from SHAP
14	0	0	0
789	1	1	1(but with probability 0.42 slightly higher than 0.35)
250	2	2	2

Table 13. Correct classification of ground truth label and predicted label for different Instances.

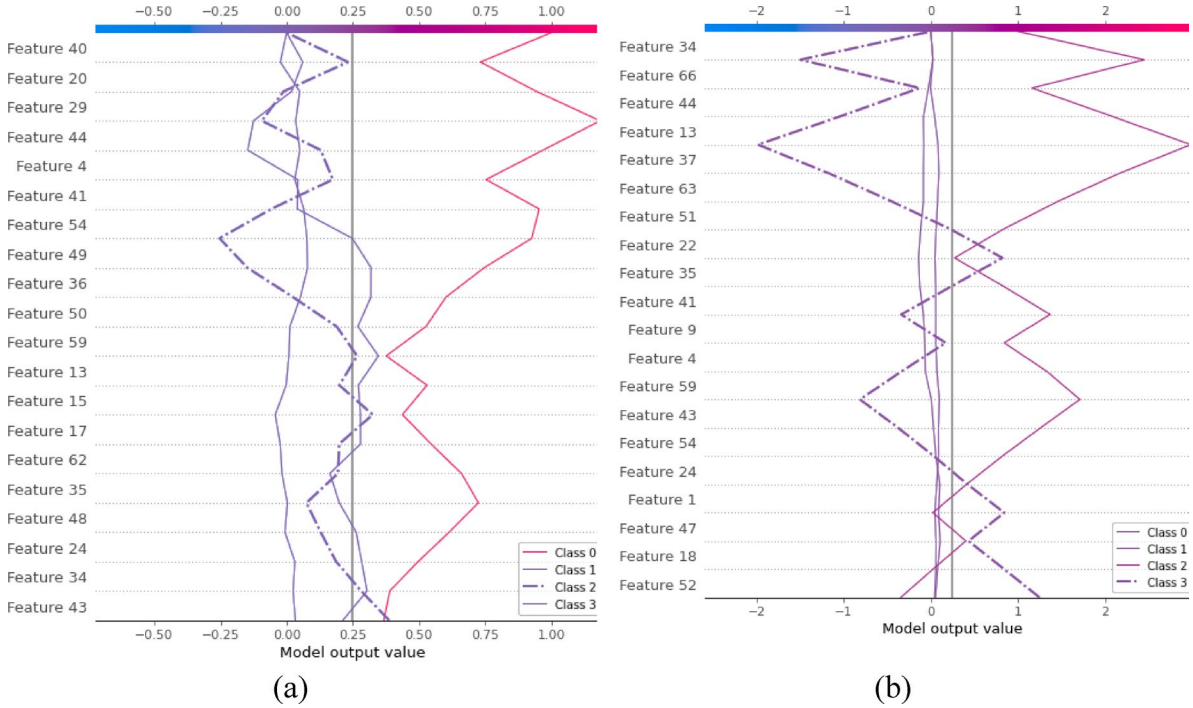


Fig. 8. Multioutput decision plot of DEAP: (**a**) For the Instance 1250, (**b**) For the Instance 30.

Observed instance	Ground truth	Predicted label from RF algorithm	Predicted label from SHAP
30	3	2	2
300	1	0	0
750	0	2	2
1250	2	0	0
21	3	3	3
1598	3	3	3

Table 14. Misclassification ground truth label and predicted label for different instances.

Case-2: misclassification
For misclassification, the dataset from DEAP experiment is considered because the cross-subject accuracy was very less in every model experimented. Here, the classes for classification are 4, LALV, LAHV, HAHV and HALV. The Fig. 8a shows multioutput decision plot for the instance 1250, whose true class is 2 and predicted class is class zero. The maximum number of features here are 70 as only 14 channels with 5 band features within each has been considered. It can be seen from the graph that model confidently misclassify it to class zero. Similarly, another instance 30 has also been shown in Fig. 8b, where the true class is 3 and the model predicted class is 2. The misclassification of Ground Truth label and predicted label for different instances are shown in Table 14.

Effect of selected features on accuracy

This experiment is done in order to understand how accuracy changes when the top important features from bee swarm plot (or summary plot) are selected for the model input. The pictures of bee swarm plot shown here contain only 20–30 features maximum. But this experiment was done with 60 features plot for all the three classes in SEED dataset which is not shown. The statistics of the feature distribution within each band is presentedin Table 15. From the Table 15, it can be seen that most of the energy is concentrated in gamma band.

Name of the band	Band – 1 (θ)	Band-2 (α)	Band-3(σ)	Band-4 (β)	Band-5 (γ)
Number of features	4	4	5	14	43

Table 15. Feature distribution statistics with each band.

S.No	Number of features selected	Classification accuracy with Random Forest Algorithm
1	310 (complete features)	0.75
2	70 top features from bee swarm plot	0.79
3	286(Random)	0.74
4	158(50%-Random)	0.71

Table 16. Accuracy variation with different features.

The gamma band is the discriminative feature which decides the basic emotions and the next contributor is beta band which is confirmed by the works done by the authors of[42,43]. Also, they are opposite for positive and negative emotions as confirmed by bee swarm plot discussed in Fig. 8a and b.

It is found that some features were not common among all the three classes, and hence, considering them also leads to the increase in total number of features, i.e., 51 to 70. With this the classification accuracy results are shown in Table 16. The results demonstrate that feature selection guided by XAI yields superior classification performance compared to both complete and randomly selected feature sets. Accuracy reduction with random features confirms that, performance gain is not attributed to feature reduction alone. It provides quantitative evidence that XAI-based feature ranking captures meaningful and discriminative emotional information. This also reduces the training run-time and prediction time. The reduced accuracy in case of more features may be due to complications in segregating the discriminative features for each emotion.

Diagnostic performance analysis using SHAP values

The dataset dependent attribution behavior of the model can be clearly inferred from the figures shown in 9 a, b,c and d. Figure 9a and b shows the DEAP dataset mean SHAP values distribution with respect to frequency domain features. It can be observed that, the mean SHAP value range and the concentration of the data are different for both training and dataset, and this divergence explains the reduced accuracy shown in Table 7. In contrast, the same plot for SEED shows that the distribution with testing and training are highly overlapped which confirms the cross-subject performance. The SHAP value variance plot from Figs. 10 (a) and 10(b) further confirms these observations. The variance shown by SEED features is not very random or not high and they are mostly concentrated together. However, in case of DEAP dataset the variance is abnormally high, i.e., the features are highly distributed and hence it reflects the behavior of poor performance with cross subject.

The SHAP variance distribution plot for the entire dataset shows the overall spread, which does not give clue about the inter subject variability and hence per subject SHAP variance was computed subsequently and is shown in Fig. 11. This allowed to get an insight on inter subject variability than aggregated variability across the subject.

It can be seen from Fig. 11 that the SHAP variance changes with individual subject for both SEED and DEAP, however the magnitude of variance is far less for the case of SEED compared to DEAP dataset. This provides insight into the characteristics of the dataset that DEAP dataset suffers from intersubject variability which leads to reduced accuracy with subject independent experiment setting. For further validation of the same, correlation analysis was performed and the results are tabulated in Table 17.

Each dataset demonstrates negative correlation together with statistically significant p value. However, SEED dataset shows very strong correlation with strong significant p value indicating its accuracy with intersubject experimental testing.

Ethical considerations in emotion recognition

Emotion recognition in mental health contexts raises ethical concerns related to data privacy, algorithmic bias, and potential misuse. EEG signals are sensitive and require secure handling and anonymization. Dataset-dependent behavior may introduce biased or unreliable inferences across populations. Explainable AI supports transparency and informed human oversight, reducing the risk of misinterpretation.

Actionable implications for model design and deployment

The SHAP-based analysis provides actionable guidance for improving emotion recognition models. The concentration of high SHAP values within specific frequency bands and a limited set of EEG channels suggests that future models may prioritize these diagnostically salient features while downweighting unstable or low-contributing inputs, thereby improving robustness and efficiency. Conversely, the dispersed and high-variance SHAP patterns observed in non-performing and cross-dataset scenarios indicate the need for architectures that better capture subject-invariant and temporal dynamics. From a deployment perspective, SHAP-based attribution stability can serve as a validation criterion prior to real-world use, while lightweight and on-device XAI implementations can support transparent and privacy-preserving deployment in mental health–related applications.

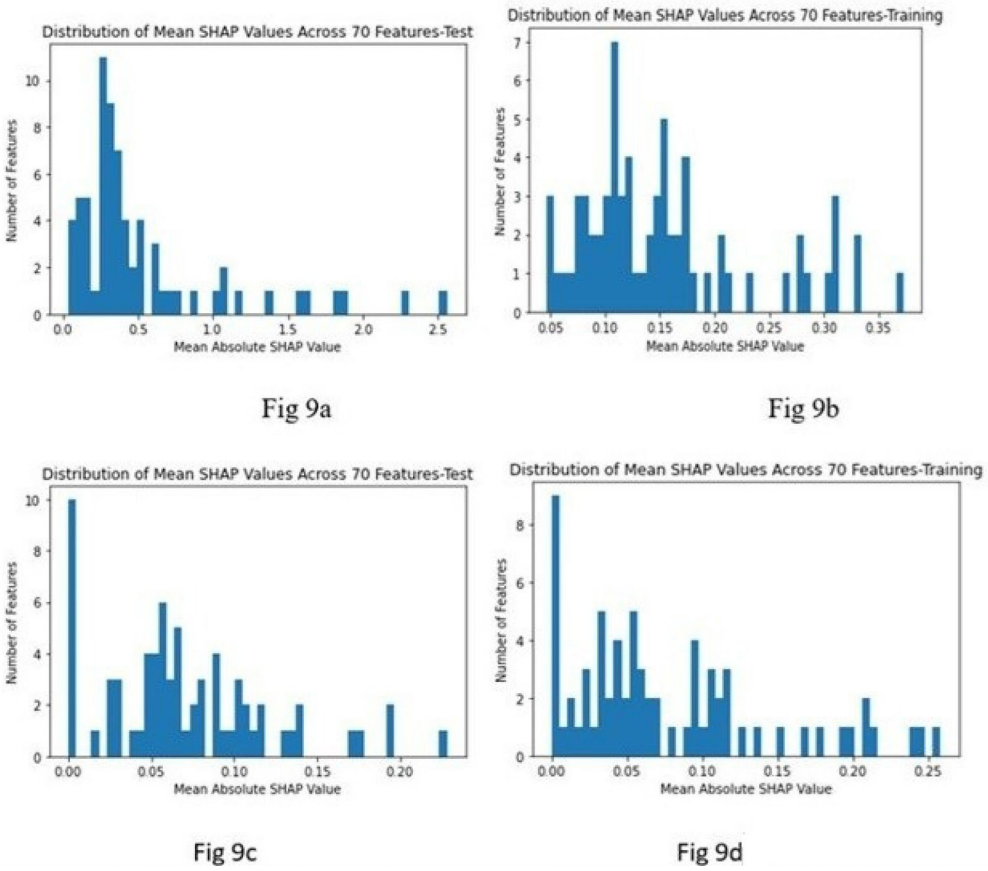


Fig. 9. Distribution of mean SHAP values for DEAP and SEED datasets (training and test sets).

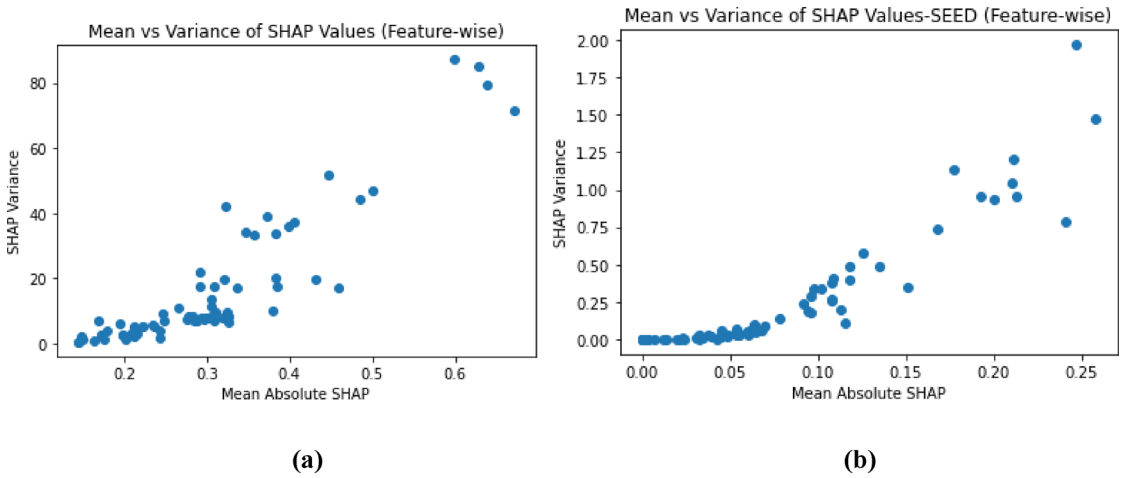
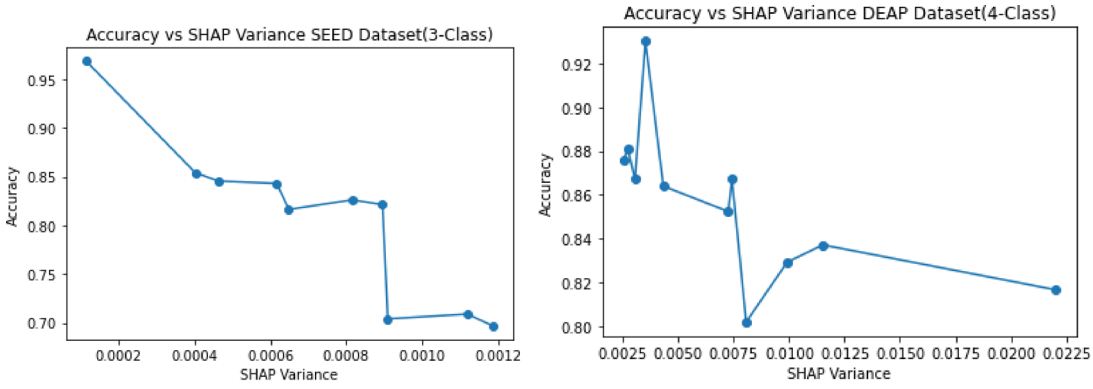


Fig. 10. SHAP variance plot (**a**) DEAP dataset (**b**) SEED Dataset.

Limitations and Future Study

SHAP The SHAP-based analysis in this study is limited to post hoc interpretability and does not contribute to model optimization. The explanations given are model dependent influenced by the model decision boundaries. Future work can involve the integration of frequency weighting based on SHAP based features.

Poor generalization This study is limited to extensive SHAP analysis to get insights into the model, rather than model optimization for better performance with subject independent accuracy. The work uses the basic features like alpha, beta, gamma, whereas the non-linearity in EEG signals can be better handled with attention based deep wavelet al.gorithms as discussed in[44]. Future work will incorporate this self-attention mechanism for subject independent accuracy improvement. Unlike post hoc methods, model-driven interpretability approaches rely



(a) (b)

Fig. 11. Individual SHAP variance plot for (**a**) SEED and (**b**) DEAP dataset.

Dataset	Correlation	*P* value
SEED	-0.9183	0.00017
DEAP	−0.6613	0.0267

Table 17. Correlation between SHAP variance and test accuracy.

on domain-specific knowledge, such as functional connectivity and latent factor representations[45], and require careful model design. While these models offer inherent interpretability compared to SHAP, their applicability to complex architectures is still evolving.

Lack of real-time data The current study has been done with publicly available data which limits the testing of real-time data. The practical feasibility of testing in real-time even in subject dependent scenario needs to be validated. The findings of this study have potential implications in real-world domains such as mental health monitoring, affective computing, and human–computer interaction. However, the challenges mentioned here needs to be addressed before deployment in clinical or real time environment. Future work will focus on bridging this gap by incorporating more diverse datasets and exploring clinically validated evaluation protocols.

Conclusions

The study elaborates a deep insight about the role of explainable AI in interpreting the model's behavior not only with the classification accuracy but also through the contribution of features using SHAP. The experimental results differentiate well performing models with compact and dense distribution and poor performing models with wide and more diffuse spread which reflects uncertainty in feature attribution. In SEED dataset, the model decisions are more predominant using the 70 features out of 310 which lies in Gamma band followed by Beta band and they are in agreement with neurological findings. Further, these dominant features were localized to a limited subset of EEG channels which indicates that the model decisions are driven by a small set of diagnostically salient neural patterns instead of uniformly distributed inputs.

The SHAP-based analysis mitigates the inherent black-box nature of deep learning models by bringing out the feature- and channel-specific decision pathways, which improves the neurophysiological interpretability for diagnosis. However, the observed variability in SHAP attributions across trials and subjects indicates instability in feature relevance which highlights the challenge of achieving consistent and subject-independent interpretations. Though the CNN and EEGNet models capture discriminative patterns, they appear limited in resolving subtle emotional boundary ambiguities, as reflected by overlapping and fluctuating feature attributions in non-performing scenarios.

In contrast, the DEAP dataset did not yield clearly dominant features which suggest that the subjective self-assessment labels introduced ambiguity in emotion representation, thereby affecting both model performance and interpretability. Although classification accuracy improved when restricted to a reduced feature subset, this work does not give a detailed investigation of feature variation dynamics, since the primary focus is on evaluating model transparency through explainable AI. In conclusion, the findings emphasize that XAI enhances interpretability and diagnostic reasoning, but the challenges related to attribution stability, neurophysiological consistency, and emotion boundary definition remain open and warrant further investigation.

Data availability

The dataset employed in this study was obtained from the link provided below. The detailed procedure to access the dataset is also available at the same source. Available with restricted access. The data are, however, available

upon request.[https://www.eecs.qmul.ac.uk/mmv/datasets/deap/download.html](https:/www.eecs.qmul.ac.uk/mmv/datasets/deap/download.html).

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Author contributions

J.M.M. and M.B.N.M. designed the experiments and analyzed the data, and wrote the main manuscript text. M.S. performed the model training, and A.S.K.P. reviewed the manuscript. All authors approved the final version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to N.M.M.B.

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